Function	Domain	Range	Mono-tonicity	Relationships	Function Derivative	Inverse	Inverse Derivative
sin x (odd)	$\mathbb{R}$	[-1,1]		$\cos^2 x + \sin^2 x = 1$ $\sin(2x) = 2 \sin x \cos x$	cos x		
sinh x (odd)	$\mathbb{R}$	$\mathbb{R}$					
cos x (even)	$\mathbb{R}$	[-1,1]		$\cos^2 x + \sin^2 x = 1$ $\cos(2x) = \cos^2 x - \sin^2 x$	-sin x		
cosh x (even)	$\mathbb{R}$	$[1,\infty)$					
tan x (odd)	$\mathbb{R} \setminus \frac{\pi}{2}\mathbb{I}_{odd}$	$\mathbb{R}$		$1 + \tan^2 x = \sec^2 x$	$\sec^2 x$		
tanh x (odd)	$\mathbb{R}$	(-1,1)					
sec x (even)	$\mathbb{R} \setminus \frac{\pi}{2}\mathbb{I}_{odd}$	$(-\infty,-1] \cup [1,\infty)$		$1 + \tan^2 x = \sec^2 x$	sec x tan x		
sech x (even)	$\mathbb{R}$	(0,1]					